

Section A — Concept Application

1) Explain the process of selecting a compatible motherboard, CPU, RAM, and storage device for a new office workstation, and justify how you would confirm they will work together.

1. Select the CPU

First decide the workstation's workload—such as Microsoft Office, web browsing, accounting software, programming, or multitasking.

Check:

- CPU socket type, such as LGA1700, LGA1851, or AM5.
- Required chipset compatibility.
- CPU power requirements.
- Whether integrated graphics are available if a separate graphics card is not required.

2. Select a compatible motherboard

Choose a motherboard specifically supporting the selected CPU.

Check:

- CPU socket matches the processor.
- Chipset supports the CPU generation/model.
- BIOS version supports the CPU.
- RAM type supported, such as DDR4 or DDR5.
- Number and type of RAM slots.
- Storage interfaces, such as M.2 NVMe and SATA.
- Appropriate motherboard size/form factor, such as ATX or Micro-ATX.

3. Select RAM

Choose RAM based on the motherboard's specifications.

For a typical office workstation:

- 16 GB is a good starting point.
- 32 GB is preferable for heavy multitasking.
- Match the motherboard's supported RAM generation (DDR4/DDR5).
- Check the supported speed and maximum capacity.
- Prefer matched modules, such as 2 × 8 GB or 2 × 16 GB, when dual-channel operation is supported.

4. Select storage

For a new workstation, an NVMe M.2 SSD is generally a good choice because it provides fast boot and application loading.

Check:

- Motherboard has a compatible M.2 slot.
- SSD interface/protocol is supported, such as PCIe/NVMe.
- Available PCIe generation and bandwidth.
- Required capacity—for example, 500 GB or 1 TB for a normal office workstation.

5. Confirm that everything works together

Before purchasing, I would create a compatibility checklist:

Component	Compatibility to verify
CPU	Socket + chipset + BIOS support
Motherboard	CPU support + RAM type + M.2/SATA support
RAM	DDR generation + speed + capacity + motherboard support
SSD	M.2 size + NVMe/PCIe support
Power supply	Required wattage + motherboard/CPU connectors
Case	Motherboard form factor + physical clearance

2) Explain the differences between HDD, SSD, and NVMe storage, and recommend the most suitable option for this workload, giving technical reasons for your choice.

Feature	HDD	SATA SSD	NVMe SSD
Technology	Magnetic spinning disks	Flash memory	Flash memory
Interface	SATA	SATA	PCIe
Typical speed	~80–160 MB/s	~450–550 MB/s	~2,000–7,000+ MB/s
Moving parts	Yes	No	No
Boot/application loading	Slow	Fast	Very fast
Noise	Some noise	Silent	Silent
Power consumption	Higher	Lower	Low–moderate
Durability	Lower due to moving parts	Higher	Higher
Cost per GB	Lowest	Moderate	Higher

3) Explain the role of the GPU and the cooling system in maintaining performance and stability under this kind of sustained load.

1. GPU (Graphics Processing Unit)

The GPU is responsible for processing and displaying graphics.

- Integrated GPU: Usually sufficient for normal office applications, web browsing, presentations, and multiple-monitor setups.
- Dedicated GPU: Useful for CAD, 3D modelling, video editing, AI workloads, or other graphics-intensive applications.
- A capable GPU reduces the graphics-processing workload placed on the CPU.
- Proper GPU cooling prevents the GPU from overheating and reducing its clock speed through thermal throttling.

2. Cooling System

- CPU heatsink and fan or liquid cooler to remove heat from the processor.
- Case fans to bring in cool air and exhaust hot air.
- Thermal paste to improve heat transfer between the CPU and heatsink.
- Proper airflow management to prevent hot air from accumulating inside the case.

4) Explain why hardware compatibility matters when assembling a computer, and identify potential issues that could arise from incompatible components

Incompatible Components	Possible Problem
CPU and motherboard socket don't match	CPU cannot be installed
CPU not supported by motherboard BIOS	Computer may not boot
DDR4 RAM installed on DDR5 motherboard	RAM cannot be installed or used
RAM speed/voltage not supported	Instability, crashes, or reduced performance
M.2 SSD not supported by motherboard	SSD may not be detected
GPU too large for the case	GPU cannot physically fit
PSU has insufficient wattage	Shutdowns, crashes, or unstable operation
Incorrect PSU connectors	Component may not receive power
Motherboard doesn't fit the case	Physical assembly is impossible

6) explain the checks you would perform to confirm the system passes POST successfully, and what you would do if it did not.

Checks to Confirm the System Passes POST

POST (Power-On Self-Test) is the initial hardware check performed by the motherboard/firmware when the computer is powered on. I would perform the following checks:

1. Power-on check
 - Press the power button and confirm the PSU, CPU, and case fans start.
 - Check that the motherboard receives power.
2. Display check
 - Confirm the monitor turns on and displays the manufacturer's logo or BIOS/UEFI screen.
 - Make sure the monitor cable is connected to the correct video output.
3. POST indicators
 - Check the motherboard's debug LEDs, POST code display, or beep codes.
 - A successful POST should not show a persistent CPU, DRAM, VGA, or BOOT error indicator.
4. BIOS/UEFI hardware detection
 - Enter BIOS/UEFI using the appropriate key, such as Delete or F2.
 - Verify that the CPU, total RAM, SSD/HDD, and other essential hardware are detected correctly.
5. Memory check
 - Confirm the installed RAM capacity is correctly reported.
 - If available, run a basic memory diagnostic after the system boots.
6. Storage and boot check
 - Confirm the SSD/HDD is detected.
 - Check that the correct boot device is selected.

Section B — Practical Coding Tasks



